

T.C.
İSTANBUL KÜLTÜR UNIVERSITY
FACULTY OF ENGINEERING
DEPARTMENT OF
ELECTRICAL & ELECTRONICS ENGINEERING



Effi-Home: Smart Energy Optimization

OZAN EMRE TUNCA

Smart Home and Energy Management System Project Report

Introduction

The increasing global demand for energy necessitates innovative approaches to manage and optimize household energy consumption. This project focuses on implementing a Smart Home and Energy Management (SH&EM) system designed to enhance energy efficiency and reduce costs. The system uses a smart algorithm to prioritize electrical loads, shift high consumption periods, and visualize load profiles for efficient energy usage.

The primary objectives of this project include:

1. Categorizing household appliances based on their energy usage.
2. Preserving household comfort while shifting energy loads during peak hours.
3. Analyzing energy consumption for typical winter and summer days.
4. Comparing energy costs with and without load shifting using a three-tiered tariff structure.

Implemented using Python, this system offers clear data analysis, visualization, and effective cost-saving solutions for smart home energy management.

Additionally, the project contributes to sustainability by demonstrating the potential to optimize energy costs. By prioritizing device groups and employing innovative algorithms, it reduces the load on energy grids while creating long-term savings for households.

Methodology

Step 1: Categorizing Household Appliances and Data

Household appliances were classified into five priority levels based on importance and energy consumption:

- Priority 1 (High Priority): Devices requiring continuous operation (e.g., Refrigerator, Internet Modem).
- Priority 2 (Medium Priority): Devices with moderate flexibility (e.g., Oven, Coffee Machine).
- Priority 3 (Low Priority): Devices with minimal urgency (e.g., Game Console, Computer).

Step 2: Defining Usage Hours

Each device's operating hours were determined based on typical winter day usage patterns, considering user habits and operational requirements. An Excel table was created to list these usage patterns and hourly energy consumption.

Additionally, detailed insights into the impact of device groups on energy efficiency were drawn. This analysis integrates load-shifting algorithms to reduce grid load during peak hours.

Step 3: Establishing Threshold Values

To optimize energy consumption, a threshold of 7500W was set as the maximum allowable power during peak hours. Any consumption exceeding this limit was redistributed through load-shifting techniques. This optimization considered individual devices and overall consumption patterns.

Step 4: SH&EM Algorithm

Our SH&EM algorithm is designed to optimize home energy management and reduce energy costs. Developed in Python, it incorporates the following steps:

- 1. Defining Threshold Values:** Establishing a threshold (7500W) based on daily energy consumption profiles.
- 2. Analyzing Peak Hours:** Identifying time intervals where total energy consumption exceeds the threshold.
- 3. Shifting Prioritized Loads:** Relocating low-priority loads to off-peak hours while maintaining user comfort. This involves rescheduling devices based on energy tariffs.
- 4. Recalculating Consumption and Costs:** Reassessing energy consumption and costs for original and shifted load profiles. Tariffs are applied to calculate total savings.
- 5. Optimization and Visualization:** Graphically visualizing the new load profiles to demonstrate their impact on the grid and cost savings.

HOUSEHOLD APPLIANCES AND PRIORITY GROUP /USAGE HOURS AND POWER	Refrigerator (High)	Oven (Medium)	LED Lighting (High)	Washing Machine (Low)	Dishwasher (Low)	Internet Modem (High)	Air Conditioner (Low)	Television (Medium)	Computer and Monitor (Low)	Vacuum Cleaner (Medium)	Electric Heater (High)	Game Console (Low)	Coffee Machine (Medium)	Electric Stove (High)	Chargers (Medium)	Music System (Low)	Air Fryer (Medium)	Deep Freezer (High)	Energy Consumption (Watt)
00:00-01:00	600	0	0	0	0	550	0	0	0	0	0	0	0	0	150	0	0	450	1750
01:00-02:00	600	0	0	0	0	550	0	0	0	0	0	0	0	0	150	0	0	450	1750
02:00-03:00	600	0	0	0	0	550	0	0	0	0	0	0	0	0	150	0	0	450	1750
03:00-04:00	600	0	0	0	0	550	0	0	0	0	0	0	0	0	150	0	0	450	1750
04:00-05:00	600	0	0	0	0	550	0	0	0	0	0	0	0	0	150	0	0	450	1750
05:00-06:00	600	0	0	0	0	550	0	0	0	0	0	0	0	0	150	0	0	450	1750
06:00-07:00	600	0	0	0	0	550	0	0	0	0	0	0	0	0	150	0	0	450	1750
07:00-08:00	600	0	350	0	0	550	0	0	0	0	0	0	1000	0	0	0	0	450	2950
08:00-09:00	600	0	350	0	0	550	0	0	0	0	0	0	0	0	0	0	1500	450	3450
09:00-10:00	600	0	0	0	0	550	0	0	0	0	0	0	0	0	0	0	0	450	1600
10:00-11:00	600	0	0	1250	0	550	0	0	0	850	0	0	0	0	0	500	1500	450	5700
11:00-12:00	600	0	0	1250	0	550	0	0	0	850	0	0	0	0	0	500	0	450	4200
12:00-13:00	600	0	0	0	0	550	0	900	0	0	0	0	0	0	0	0	0	450	2500
13:00-14:00	600	0	0	0	0	550	0	0	0	0	0	0	1000	0	0	0	0	450	2600
14:00-15:00	600	0	0	0	0	550	0	0	0	0	0	0	0	0	0	0	0	450	1600
15:00-16:00	600	0	0	0	0	550	0	0	0	0	1500	950	0	0	0	500	0	450	4550
16:00-17:00	600	2800	0	0	0	550	0	0	0	0	1500	950	0	0	0	500	0	450	7350
17:00-18:00	600	2800	350	0	0	550	0	900	0	0	1500	0	0	0	0	500	0	450	7650
18:00-19:00	600	0	350	0	0	550	0	900	1300	0	1500	0	0	2000	0	500	0	450	8150
19:00-20:00	600	0	350	0	2500	550	0	900	1300	0	1500	0	0	2000	0	0	0	450	10150
20:00-21:00	600	0	350	0	2500	550	0	900	1300	850	1500	0	0	0	0	0	0	450	9000
21:00-22:00	600	0	350	0	0	550	0	900	0	0	1500	0	0	0	0	0	0	450	4350
22:00-23:00	600	0	350	0	0	550	0	900	0	0	1500	0	0	0	0	0	0	450	4350
23:00-24:00	600	0	0	0	0	550	0	0	0	0	0	0	0	0	0	0	0	450	1600

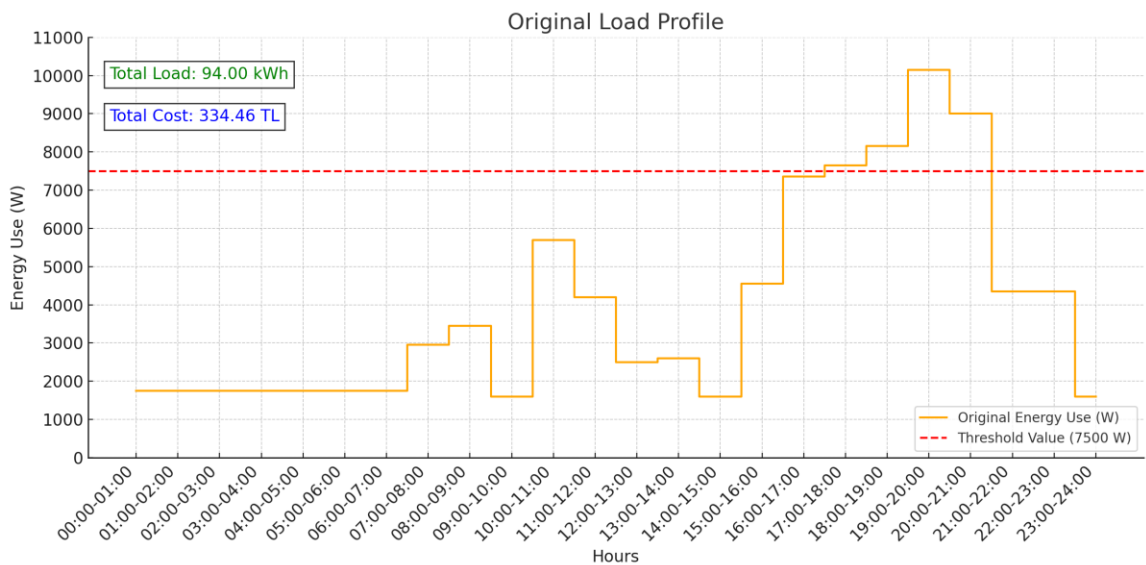
Sample Data

Step 5: Visualization and Comparison

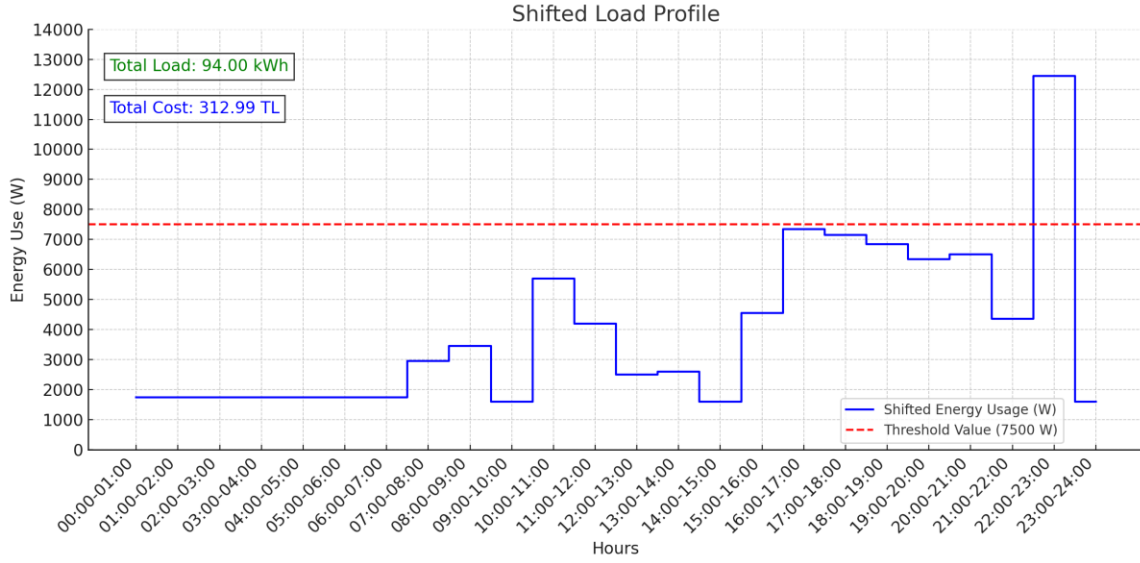
We created a sample data to observe the working principle of this algorithm:

The following graphical outputs were generated:

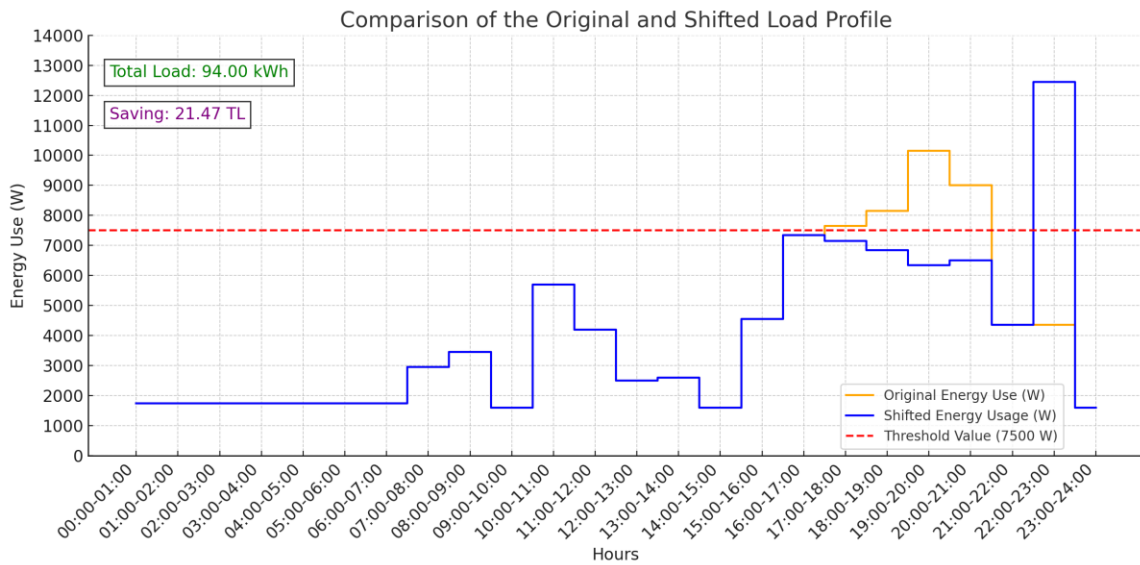
- 1. **Original Load Profile:** Illustrates energy consumption without load shifting.
- 2. **Shifted Load Profile:** Displays optimized consumption after applying the load shifting algorithm.
- 3. **Comparison Graph:** Highlights savings achieved through load shifting.



Original Load Profile



Shifted Load Profile



Comparison Graph

These graphs reveal the substantial impact of the optimization process, aligning with the threshold values for grid efficiency.

Step 6: Calculating Energy Costs

Energy costs for both scenarios were calculated using a three-tiered tariff system:

Original Scenario: Higher consumption during peak hours resulted in higher costs.

Shifted Scenario: Load shifting reduced peak-hour consumption, leading to cost savings.

For example, the average monthly household energy consumption is 430 kWh, costing approximately 1000 TL. With this system, annual savings of 1266,55 TL can be achieved based on daily savings of 3,47 TL.

Detailed Calculation of Energy Costs:

Monthly Energy Consumption:

The average monthly energy consumption is 2820 kWh. (For Project Data)

Daily Savings from Load Shifting:

Load shifting results in 21.47 TL savings per day

Monthly Savings:

If 21,47 TL is saved each day, the total monthly savings can be calculated as:

$$21,47 \text{ TL/day} \times 30 \text{ days} = 641,1 \text{ TL/month}$$

Annual Savings:

Based on daily savings of 21.47 TL, the annual savings would be:

$$21,47 \text{ TL/day} \times 365 \text{ days} = 7836,55 \text{ TL/year}$$

Energy Cost Comparison:

Energy Cost Without Load Shifting: 334,46 TL

Energy Cost With Load Shifting: 312,99 TL

The daily savings from load shifting can be calculated as:

$$334,46 \text{ TL} - 312,99 \text{ TL} = 21,47 \text{ TL/day}$$

Results and Analysis

Key Observations

- Total Load: Remains constant at 94,00 kWh for both scenarios.
- Energy Cost Without Load Shifting: 334,46 TL.
- Energy Cost With Load Shifting: 312,99 TL.
- Daily Savings: 21,47 TL.
- Annual Savings Potential: Approximately 7836,55 TL, based on daily savings.

Graphical Analysis

- 1. Original Load Profile:** Shows peak-hour surges exceeding the 7500W threshold.
- 2. Shifted Load Profile:** Balanced energy consumption eliminates threshold breaches.
- 3. Comparison Graph:** Demonstrates significant improvements in energy distribution and cost reductions.

The graphs highlight the dual benefits of cost reductions and reduced grid load.

Insights

The load-shifting algorithm effectively reduces energy costs while preserving household comfort. Priority-based scheduling ensures uninterrupted operation of critical devices. Furthermore, the seasonal differences in usage patterns provide valuable insights for future optimization efforts.

Conclusion

The Smart Home and Energy Management System demonstrates significant energy-saving potential through intelligent load management. Key benefits include:

1. Reducing energy consumption during peak hours.
2. Enhancing cost efficiency with minimal disruption to user comfort.
3. Scalability to accommodate seasonal changes (e.g., summer analysis forthcoming).

Future improvements may involve integrating renewable energy sources and real-time user feedback for enhanced optimization. This project highlights the practical application of Python in addressing real-world energy management challenges, offering scalable and user-friendly solutions for modern smart homes.

References

<https://www.tedas.gov.tr/A/1/elektrik-tarifeleri/RoutePage/63c65a04d27de36b22f9cf1f>

Enerji Piyasası Düzenleme Kurumu (EPDK). (2024). "Elektrik Tarifeleri ve Maliyet Analizi." <https://epdk.gov.tr/Detay/Icerik/3-0-1-3/elektriktarifeler>

Python Software Foundation. (2024). Python Language Reference, version 3.10. <https://www.python.org>

Kluyver, T., et al. (2016). "Jupyter Notebooks - A Publishing Format for Reproducible Computational Workflows." In F. Loizides and B. Schmidt (Eds.), Positioning and Power in Academic Publishing: Players, Agents and Agendas (pp. 87-90). IOS Press. <https://jupyter.org>

